

R Package Validation Report — Tier 1 Screening

digest 0.6.39

Package digest
Version 0.6.39 [EV-002]
Origin cran [EV-009]
Tarball sha256 8bf048b49b2d17077138fae758bda56bbd53278d9437f2fdeaedf979c90a13c9 [EV-008]
Intended-use tier gxp-support
Report generated (UTC) 2026-09-05T19:54:34Z

Decision

GO-WITH-CONDITIONS — overall risk score 0.3769 [EV-024]
 0 = low risk, 1 = high; config v1.1.0

Rules evaluated (fired or not, with inputs and thresholds):

Rule	Input	Condition	Fired
overall_score_threshold	0.3769	score $\geq$ 0.2000 / 0.4000	FIRED
news_missing	0	$\geq$ 0.5000	no
news_not_measurable	0	is_na	no
low_downloads	1069522	$\geq$ 1000	no
bugs_mostly_open	1	$\geq$ 0.5000	FIRED
no_source_control	eddelbuettel/digest	is_na	no
exports_undocumented	0	$\geq$ 0.5000	no
license_risk	(no input)	$\geq$ 0.0000	no
heavy_dependencies	0.8176	$\geq$ 0.5000	FIRED

Conditions

- Majority of recent issues unresolved — review the bug tracker before use.
- Large dependency footprint — each dependency needs its own screening decision.

Per-Metric Breakdown ({riskmetric})

Metric	Risk (0–1)	Evidence
bugs_status	1	EV-025
dependencies	0.8176	EV-026
downloads_1yr	0.9900	EV-027
export_help	0	EV-028
exported_namespace	0.6225	EV-029
has_bug_reports_url	1	EV-030
has_examples	1	EV-031
has_maintainer	1	EV-032
has_news	0	EV-033
has_source_control	1	EV-034
has_vignettes	1	EV-035

Metric	Risk (0–1)	Evidence
has_website	1	EV-036
news_current	0	EV-037
reverse_dependencies	1	EV-038
size_codebase	0.1620	EV-039

Scoring Strategy & Category Subscores

Category	Weight	Subscore (risk)	(0–1)	Member metrics
code	0.5000	0.5340		dependencies: 0.8176 [EV-026], exported_namespace: 0.6225 [EV-029], remote_checks: NA, size_codebase: 0.1620 [EV-039]
documentation	0.1500	0.5000		export_help: 0 [EV-028], has_examples: 1 [EV-031], has_news: 0 [EV-033], has_vignettes: 1 [EV-035], has_website: 1 [EV-036], license: NA, news_current: 0 [EV-037]
maintenance	0.2000	1.0000		bugs_status: 1 [EV-025], has_bug_reports_url: 1 [EV-030], has_maintainer: 1 [EV-032], has_source_control: 1 [EV-034]
popularity	0.1500	0.9950		downloads_1yr: 0.9900 [EV-027], reverse_dependencies: 1 [EV-038]

Rubric: each category subscore is the equal-weighted mean of its member metrics' non-null riskmetric scores (0 = low risk, 1 = high risk); weights disclose the relative importance of each category. Category subscores are display-only — the overall score and the decision thresholds are unchanged.

Classification & Sign-off

Not classified — no intended use or justification was recorded for this assessment.

Risk classification: PENDING HUMAN REVIEW — this tool does not classify.

Sign-off: PENDING HUMAN REVIEW — machine-generated screening output; human QA sign-off pending. This tool never signs.

Evidence Register

ID	Metric	Value	Mode
EV-001	description.package	digest	local
EV-002	description.version	0.6.39	local
EV-003	description.title	Create Compact Hash Digests of R Objects	local
EV-004	description.license	GPL (≥ 2)	local
EV-005	description.maintainer	Dirk Eddelbuettel jedd@debian.org	local
EV-006	description.url	https://github.com/eddelbuettel/digest , https://eddelbuet...	local
EV-007	description.bug_reports	https://github.com/eddelbuettel/digest/issues	local
EV-008	tarball.sha256	8bf048b49b2d17077138fae758bda56bbd53278cd91437f2fdeaedf979c...	local
EV-009	ingest.origin	cran	local
EV-010	ingest.source_url	https://cran.r-project.org/src/contrib/digest_0.6.39.tar.gz	local
EV-011	cran.version	0.6.39	network

ID	Metric	Value	Mode
EV-012	cran.title	Create Compact Hash Digests of R Ob- jects	network
EV-013	cran.publication_date	2025-11-19	network
EV-014	cran.maintainer	Dirk Eddelbuettel jedd@debian.org	network
EV-015	cran.license	GPL ($i=2$)	network
EV-016	cran.url	https://github.com/eddelbuettel/digest, https://eddelbuet...	network
EV-017	cran.imports_count	1	network
EV-018	downloads.last_month_total	1069522	network
EV-019	github.repo	eddelbuettel/digest	network
EV-020	github.stars	122	network
EV-021	github.forks	50	network
EV-022	github.open_issues	6	network
EV-023	github.last_push	2026-08-21T11:31:45Z	network
EV-024	riskmetric.overall_score	0.3769	riskmetric
EV-025	riskmetric.score.bugs_status	1	riskmetric
EV-026	riskmetric.score.dependencies	0.8176	riskmetric
EV-027	riskmetric.score.downloads_1yr	0.9900	riskmetric
EV-028	riskmetric.score.export_help	0	riskmetric
EV-029	riskmetric.score.exported_namespace	0.6225	riskmetric
EV-030	riskmetric.score.has_bug_reports_url	1	riskmetric
EV-031	riskmetric.score.has_examples	1	riskmetric
EV-032	riskmetric.score.has_maintainer	1	riskmetric
EV-033	riskmetric.score.has_news	0	riskmetric
EV-034	riskmetric.score.has_source_control	1	riskmetric
EV-035	riskmetric.score.has_vignettes	1	riskmetric
EV-036	riskmetric.score.has_website	1	riskmetric
EV-037	riskmetric.score.news_current	0	riskmetric
EV-038	riskmetric.score.reverse_dependencies	1	riskmetric
EV-039	riskmetric.score.size_codebase	0.1620	riskmetric
EV-040	riskmetric.raw.downloads_1yr	14810586	riskmetric
EV-041	riskmetric.raw.export_help	FALSE	riskmetric
EV-042	riskmetric.raw.has_bug_reports_url	1	riskmetric
EV-043	riskmetric.raw.has_maintainer	Dirk Eddelbuettel jedd@debian.org	riskmetric
EV-044	riskmetric.raw.has_news	0	riskmetric
EV-045	riskmetric.raw.has_source_control	https://github.com/eddelbuettel/digest	riskmetric
EV-046	riskmetric.raw.has_vignettes	3	riskmetric
EV-047	riskmetric.raw.license	GPL ($i=2$)	riskmetric
EV-048	riskmetric.raw.size_codebase	776	riskmetric
EV-049	riskmetric.excluded_metrics	assess_covr_coverage, sess_r_cmd_check	as- riskmetric

Tier-2 Sandbox Validation

Deep validation in a throwaway R sandbox (package code executed in isolation): R CMD check, the package's own test suite and test coverage, pinned to the recorded CRAN snapshot.

Sandbox status	ok
Tests (passed/total)	0 / 0 (failed 0, skipped 0)
R CMD check	warnings (errors 0, warnings 1, notes 1)
Test coverage	0.15%
R version (sandbox)	R version 4.5.3 (2026-03-11)
Platform (sandbox)	x86_64-pc-linux-gnu
Sandbox image	rval-sandbox:0.1.0
CRAN snapshot	2026-04-15
Duration	20.75 s
Finished at (UTC)	2026-09-05T19:53:48Z

IQ — environment and spec fingerprint

R version (sandbox)	R version 4.5.3 (2026-03-11)
Platform (sandbox)	x86_64-pc-linux-gnu
Sandbox image	rval-sandbox:0.1.0
Spec sha256	d9739cb5e7be9134640afa4da0469536e95edf6096ec10f0f29d83998e8e3c97
Spec config_version	0.1.0

OQ — requirement execution

Validation status: **pass** — cases pass 6, diff 0, fail 0, error 0.

Requirement	Risk	Linked cases	Status
REQ-001: MD5, SHA-1, SHA-256 and CRC-32 of known inputs reproduce the published algorithm test vectors byte-for-byte (RFC 1321...	high	GM-001, GM-002, GM-003, GM-004, GM-005	pass
REQ-002: Atypical or malformed inputs degrade gracefully: an error or an empty/NA result is acceptable; a crash or a silently ...	medium	F, Z, -, 0, 0, 1	pass

PQ — intended-use scenarios

Scenario	Description	Status
PQ-001	Content-integrity hashing of analysis artifacts reproduces the algorithm reference vectors exactly.	pass

Requirement traceability matrix

REQ	Case	Type	Result	Detail
REQ-001	GM-001	golden	pass	identical
REQ-001	GM-002	golden	pass	identical
REQ-001	GM-003	golden	pass	identical
REQ-001	GM-004	golden	pass	identical
REQ-001	GM-005	golden	pass	identical
REQ-002	F	—	untested	
REQ-002	Z	—	untested	
REQ-002	-	—	untested	
REQ-002	0	—	untested	
REQ-002	0	—	untested	
REQ-002	1	—	untested	
—	FZ-001	fuzz	pass	wrong_type: pass (graceful error); empty_string: pass (result visibly changed)

Mutation testing (suite defect-detection power)

Mutants	10 (killed 0, survived 10, timeout 0)	
Mutation score	0.00	Surviving mutants
Operators	const_off_by_one, relop_swap	
(behavior changes this suite failed to detect):		

Mutant	File	Line	Operator
M-001	R/AES.R	103	const_off_by_one
M-002	R/digest.R	77	relop_swap
M-003	R/sha1.R	132	const_off_by_one
M-004	R/digest.R	98	const_off_by_one
M-005	R/sha1.R	87	const_off_by_one
M-006	R/sha1.R	284	relop_swap
M-007	R/vdigest.R	166	const_off_by_one
M-008	R/sha1.R	368	relop_swap
M-009	R/AES.R	91	const_off_by_one
M-010	R/AES.R	44	const_off_by_one

automated validation verdicts are decision-support evidence and never replace qualified human review and sign-off; the mutation operator set is finite, so the mutation score is an upper-biased estimate of true defect-detection power

Annex: System Information

Generator: rval-gateway 0.7.0 (Python stdlib); R version 4.5.3 (2026-03-11); {riskmetric} 0.2.7; decision config v1.1.0 (sha256 531ab81389ab28ca59dc74ef...)
evidence collection mode: network.

Evidence collected at (UTC) 2026-09-05T19:47:28Z
Assessment scored at (UTC) 2026-09-05T19:47:27Z
Network sources ok / failed 3 / 0

Approval Timeline

Audit-chain events for this package (assessment → electronic signature → repository publish), in chain order; details carry the sha256 record-hash prefix of each tamper-evident audit record.

Event	Actor	Timestamp (UTC)	Detail / record
assessment	dev-mode	2026-09-05T19:47:28Z	GO-WITH-CONDITIONS (tier gxp-support, score 0.3769) [1f156ea72719]
tier2	dev-mode	2026-09-05T19:48:43Z	tier2 ok — tests 0/0, coverage 0.15%, check warnings — validation spec present, approval pending — smoke mode; validation section suppressed [66f8d2a90d4b]
esign	qa.reviewer	2026-09-05T19:51:34Z	signed 'approved' for digest spec v0.1.0 [badbf7f5c4a1]
tier2_validation	qa.reviewer	2026-09-05T19:54:34Z	tier2 validation pass (req 2/2, mutation 0) [42d867e08cc2]

Provenance. Generator: rval-gateway 0.7.0 (Python stdlib). Scores computed by an R subprocess: R version 4.5.3 (2026-03-11), {riskmetric} 0.2.7 (two-process pattern; code-executing metrics excluded — assessed package code was not executed). Decision config v1.1.0 (sha256 531ab81389ab). Evidence collection mode: network. Hard rule: no statement without an evidence ID; sections lacking evidence are suppressed. Tier-1 screening output under a two-tier strategy; metrics inform a qualified human's assessment and are not a proof of correctness.